DRUGS AFFECTING THE ENDOCRINE SYSTEM

Classification and Prototype

1. ANTIDIABETICS

Classification

By mechanism of action
By route of administration

A. Insulins

Classification by Onset / Duration

Rapid-acting

Insulin lispro
Insulin aspart
Insulin glulisine

Short-acting

Regular insulin

Intermediate-acting

NPH insulin

Long-acting

Insulin glargine
Insulin detemir
Insulin degludec

Premixed

Short/intermediate combinations
Rapid/intermediate combinations

Prototype: Regular Insulin

MOA:

Binds insulin receptors; increases cellular glucose uptake; inhibits hepatic glucose production

Use:

Acute hyperglycemia; DKA; perioperative glucose control

Key ADR:

Hypoglycemia; hypokalemia

B. Oral and Non-Insulin Injectable Agents

Classification by Mechanism

Biguanides: Metformin
Sulfonylureas: Glipizide, glyburide, glimepiride

Meglitinides: Repaglinide, nateglinide

Thiazolidinediones (TZDs): Pioglitazone, rosiglitazone

DPP-4 Inhibitors: Sitagliptin, saxagliptin, linagliptin

GLP-1 Receptor Agonists: Exenatide, liraglutide, dulaglutide, semaglutide

SGLT2 Inhibitors: Canagliflozin, dapagliflozin, empagliflozin

Alpha-Glucosidase Inhibitors: Acarbose, miglitol

Prototype: Metformin

Class:

Biguanide

MOA:

Decreases hepatic glucose production; decreases intestinal glucose absorption; improves insulin sensitivity

Use:

First-line therapy for Type 2 Diabetes Mellitus

Key ADR:

Gastrointestinal upset; lactic acidosis (rare, serious)

2. HORMONES

Classification

By site of origin

By primary function

Excludes corticosteroids and insulin

A. Thyroid Hormones

Class

Thyroid agents

Prototype: Levothyroxine

MOA:

Synthetic T4; converted to T3 peripherally; binds nuclear receptors; regulates gene transcription; affects metabolism, growth, development

Use:

Hypothyroidism; myxedema coma; TSH suppression

Key ADR:

Hyperthyroidism signs if over-dosed (tachycardia, weight loss, heat intolerance)

B. Antithyroid Agents

Class

Thioamides

Prototype: Methimazole

MOA:

Inhibits thyroid peroxidase; blocks thyroid hormone synthesis

Use:

Hyperthyroidism (Graves' disease)

Key ADR:

Agranulocytosis; hepatotoxicity

C. Sex Hormones

Estrogens

Prototype: Estradiol

MOA:

Binds estrogen receptors; regulates gene expression

Use:

Hormone replacement therapy; contraception

Key ADR:

Thromboembolism; endometrial cancer; breast cancer

Progestins

Prototype: Progesterone

MOA:

Binds progesterone receptors; prepares endometrium for pregnancy

Use:

Contraception; hormone replacement therapy

Key ADR:

Breakthrough bleeding; mood changes

Androgens

Prototype: Testosterone

MOA:

Binds androgen receptors; promotes male secondary sex characteristics; anabolic effects

Use:

Hypogonadism; delayed puberty

Key ADR:

Virilization in women; polycythemia; hepatotoxicity

D. Pituitary and Hypothalamic Hormones

Posterior Pituitary

Prototype: Desmopressin

MOA:

Synthetic vasopressin (ADH) analog; binds V2 receptors in kidney; increases water reabsorption

Use:

Central diabetes insipidus; nocturnal enuresis

Key ADR:

Hyponatremia; water intoxication

Anterior Pituitary

Prototype: Octreotide

MOA:

Synthetic somatostatin analog; inhibits growth hormone, glucagon, insulin secretion

Use:

Acromegaly; carcinoid tumors

Key ADR:

Gallstones; GI upset; bradycardia

3. CORTICOSTEROIDS

Classification

By primary clinical use (glucocorticoid vs mineralocorticoid)

By duration of action

A. Glucocorticoids

Class

Anti-inflammatory

Immunosuppressant

Prototype: Prednisone

MOA:

Binds cytoplasmic glucocorticoid receptors; translocates to nucleus; alters gene transcription; decreases inflammatory mediators (prostaglandins, leukotrienes, cytokines); suppresses immune response

Use:

Inflammatory conditions; autoimmune conditions (asthma, rheumatoid arthritis, lupus, allergic reactions)

Key ADR:

Hyperglycemia; osteoporosis; immunosuppression; adrenal suppression; Cushing's syndrome; peptic ulcers

B. Mineralocorticoids

Class

Electrolyte regulator

Prototype: Fludrocortisone

MOA:

Binds mineralocorticoid receptors in distal tubule; promotes sodium reabsorption; promotes potassium excretion

Use:

Addison's disease (adrenal insufficiency); aldosterone replacement

Key ADR:

Hypertension; edema; hypokalemia